RAJALAKSHMI ENGINEERING COLLEGE

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Y2

Started on	Wednesday, 30 September 2026, 9:39 AM
State	Finished
Completed on	Wednesday, 30 September 2026, 9:40 AM
Time taken	38 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int findDuplicateBinarySearch(int arr[], int n) {
    int low = 1;
    int high = n - 1;

    while (low < high) {
        int mid = low + (high - low) / 2;
        int count = 0;

        // Count elements less than or equal to mid
        for (int i = 0; i < n; i++) {
            if (arr[i] <= mid) {
                count++;
            }
        }

        // If count is greater than mid, duplicate is in the lower half
        if (count > mid) {
            high = mid;
        } else {
            low = mid + 1;
        }
    }
    return low;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) return 0;

    int *arr = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    printf("%d\n", findDuplicateBinarySearch(arr, n));

    free(arr);
    return 0;
}

```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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